

Introductions

- Why are you interested in GWAS
- Level of confidence with R
 - (beginner, medium, advance)
- Level of confidence with linux and servers
 - (beginner, medium, advance)



Introduction to **GWAS**:

A quick overview

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Something you need to carefully look at, or that may impair your GWAS



Something to do, or that optimizes your GWAS



Don't. Discourage to use this.



Smart tip. Something that makes the trick.



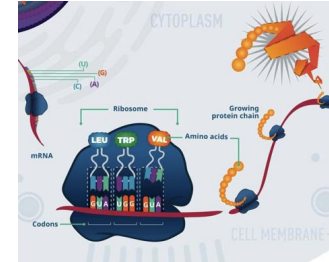
Advanced. Something to dive in.



Genetic basis of GWAS-GWP

Phenotypic variation
Genetic variants

- Allele substitution effect



- Infinitesimal model

Genetic basis of GWAS-GWP

- A study that agnostically tests hundreds of thousands of single nucleotide polymorphisms (SNPs) densely spaced across the genome for association with a given disease or trait.
- Rationale:
 - Not limited by a priori knowledge of disease process or the results of linkage studies
 - If large SNP density, each gene is expected to be in LD with at least 1 SNP
 - Preferable approach (so far) for diseases of complex etiology
 - Evidence for functional SNPs outside of coding regions

“First, our studies provide convincing evidence that the genome is pervasively transcribed, such that the majority of its bases can be found in primary transcripts, including non-protein-coding transcripts, and those that extensively overlap one another.” ENCODE, Nature June 2007



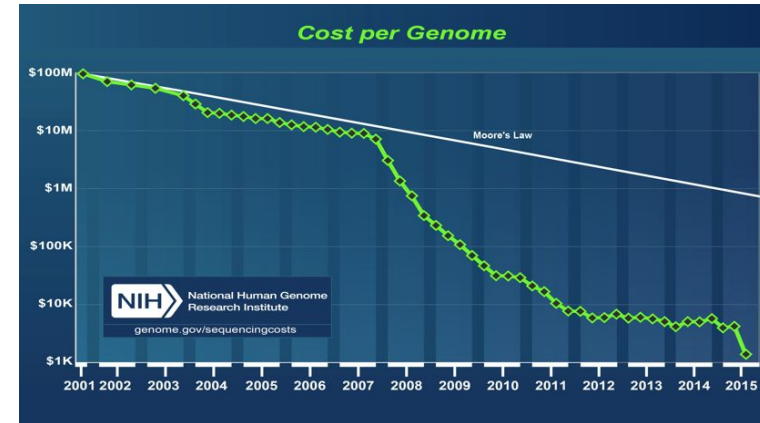
Genetic basis of GWAS-GWP

- Exposure:
 - SNPs
 - Indels
 - *Might* - copy number variants (CNV), haplotypes, other type of markers
- Outcome: easily adaptable to either dichotomous or quantitative outcomes
- Study Populations: Case-control, cross-sectional, or cohort. Family or population based.
 - Need to have large sample sizes!
- Analytic methods: logistic/linear regression
 - Flexibility for different genetic models - additive, dominant, recessive.



Genetic basis of GWAS-GWP

- Genotyping – Since early 2000 (25-100€)
 - Genome sequencing - (approx. 1000 €)
- Quantitative genetics – many genes – need many markers
- Use LD between gene and marker
- ⚠ □ We assume that all genes are in LD with at least one marker (we need a high coverage throughout the genome)



conduction a GWAS



Costs matter - approximate current **SNP array prices**

- ☐ Bacteria
- ☐ Bovine
- ☐ Canine
- ☐ Chicken
- ☐ Drosophila
- ☐ Equine
- ☐ Hamster
- ☐ Human
- ☐ Maize
- ☐ Mammalian
- ☐ Mouse
- ☐ Nematode
- ☐ Ovine
- ☐ Plant
- ☐ Porcine
- ☐ Rat
- ☐ Rice
- ☐ Shrimp
- ☐ Soybean
- ☐ Tilapia
- ☐ Tomato
- ☐ Virus
- ☐ Yeast
- ☐ Zebrafish



Microarray



For Research Use Only



DNA

BovineHD DNA Analysis Kit

This comprehensive genome-wide bovine genotyping array kit features over 777,000 SNPs, and is compatible with any breed of beef or dairy cattle.



Microarray



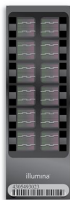
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DNA

BovineSNP50 v3 DNA Analysis BeadChip

This BeadChip microarray provides high density multi-sample bovine genotyping for genome characterization of major dairy and beef cattle breed types.



OvineSNP50 DNA Analysis Kit

This sheep microarray features over 54,241 evenly spaced SNP probes for genome-wide association studies, genome-wide selection, and genetic merit determination.[Read More...](#)

The BeadChip was developed in collaboration with leading ovine researchers from AgResearch, Baylor UCSC, CSIRO, and the USDA as part of the International Sheep Genomics Consortium. It features over 54,241 evenly spaced probes that target single nucleotide polymorphisms (SNPs).

EquineSNP50 Genotyping BeadChip

The EquineSNP50 Genotyping BeadChip features more than 54,000 evenly spaced and validated SNPs derived from the EquCab2 assembly. This 12-sample Infinium® BeadChip presents a cost-effective and high-quality genotyping solution for equine research.

Canine HD (12-sample)

170k

Featuring highly polymorphic SNP content and providing uniform genomic coverage, the CanineHD BeadChip enables the interrogation of genetic variation in any domestic dog breed. Importantly, this BeadChip presents an average of greater than 70 markers per megabase (Mb), providing ample SNP density for robust within-breed association and copy number variation (CNV) studies. This BeadChip contains more than 170,000 markers placed on the CanFam2.0 reference sequence. Illumina developed the BeadChip in collaboration with the LUPA Consortium, which includes 22 European universities and other partners such as the Broad Institute.

GGP Porcine HD Array

This genome-wide porcine genotyping array is ideal for marker-assisted selection and prediction applications. Array content includes:

- 70,000 SNPs for all major porcine breeds
- Average marker spacing of ~42 kb
- 20 key causative mutations

GGP Porcine LD Array

This array is designed for marker-assisted selection, Illumina PorcineSNP60 imputation, GGP Porcine HD imputation, and prediction applications. Array content includes:

- More than 10,000 SNPs for all major porcine breeds
- Average marker spacing of ~250 kb
- ~20 important causative mutations

Costs matter - approximate current **SNP array prices**



Microarray



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DNA

MaizeLD BeadChip Kit

Microarray kit for maize breeding applications and assessment of essentially derived varieties. Samples used include the Plant Variety Protection Act panel.



Microarray



For Research Use Only



DNA

MaizeSNP50 DNA Analysis Kit

This array enables genetic variation analysis across maize lines. It includes over 50,000 validated markers derived from the B73 corn reference sequence.



For Research Use Only



DNA



Microarray

For Research Use Only

DNA

GGP Potato Arrays 12k

Identify resistance regions with maximum coverage using the GeneSeek Genomic Profiler (GGP) potato microarray. [Read More...](#)

Select Product(s)

[What size kit do I need?](#)

0



GGP Potato-24 v4.0 (48 samples) ⓘ
20044459

[Sign in](#) to see pricing and favorite products.

0



GGP Potato-24 v4.0 (288 samples) ⓘ
20044820

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GGP Potato-24 v4.0 (1152 samples) ⓘ
20044821

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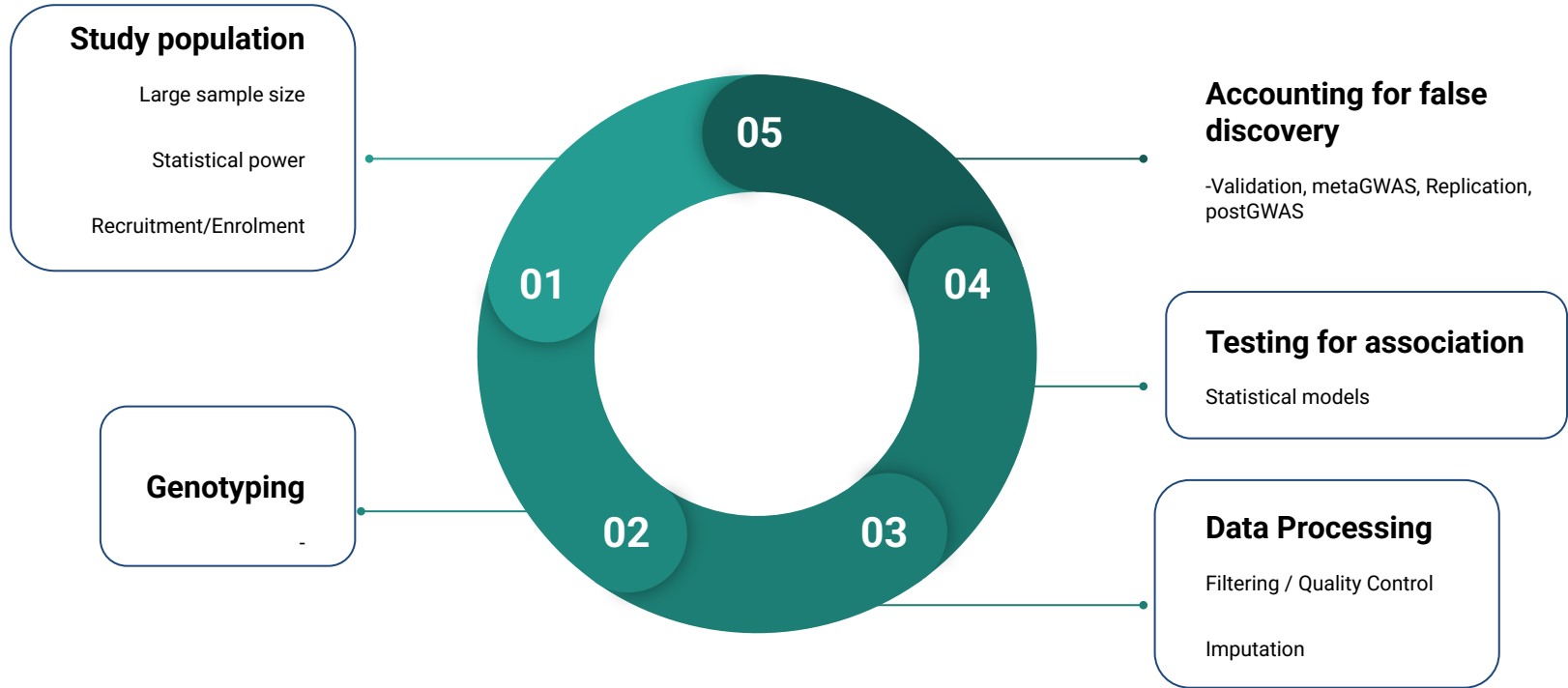
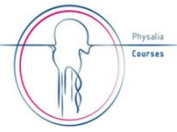
Genome-wide association studies (**GWAS**)

1. Genotyping costs have decreased dramatically over the last 10 years, in most species
2. Large number of genotyping experiments
→ explosion of GWAS experiments!

Powerful (*supposedly*) approach to the identification of genes/genomic regions involved in plant, animal and human phenotypes



conduction a GWAS



Genetic basis of GWAS-GWP

Phenotypic variation
Genetic variants

$$P = G + E$$

- Continuous trait
- Categorical trait (threshold model)



Genetic basis of GWAS-GWP

Phenotypic variation
Genetic variants
Linkage disequilibrium

$$P = G + E$$

Baby model $y = \mu + x\beta + \epsilon$

Genotype-Phenotype association → causal or functional link



Genome-wide association studies (**GWAS**)

- Based on linear regression models (mathematics, linear algebra, statistics)
- Uses linkage disequilibrium between genomic markers and genes

Genome-wide association studies (**GWAS**)

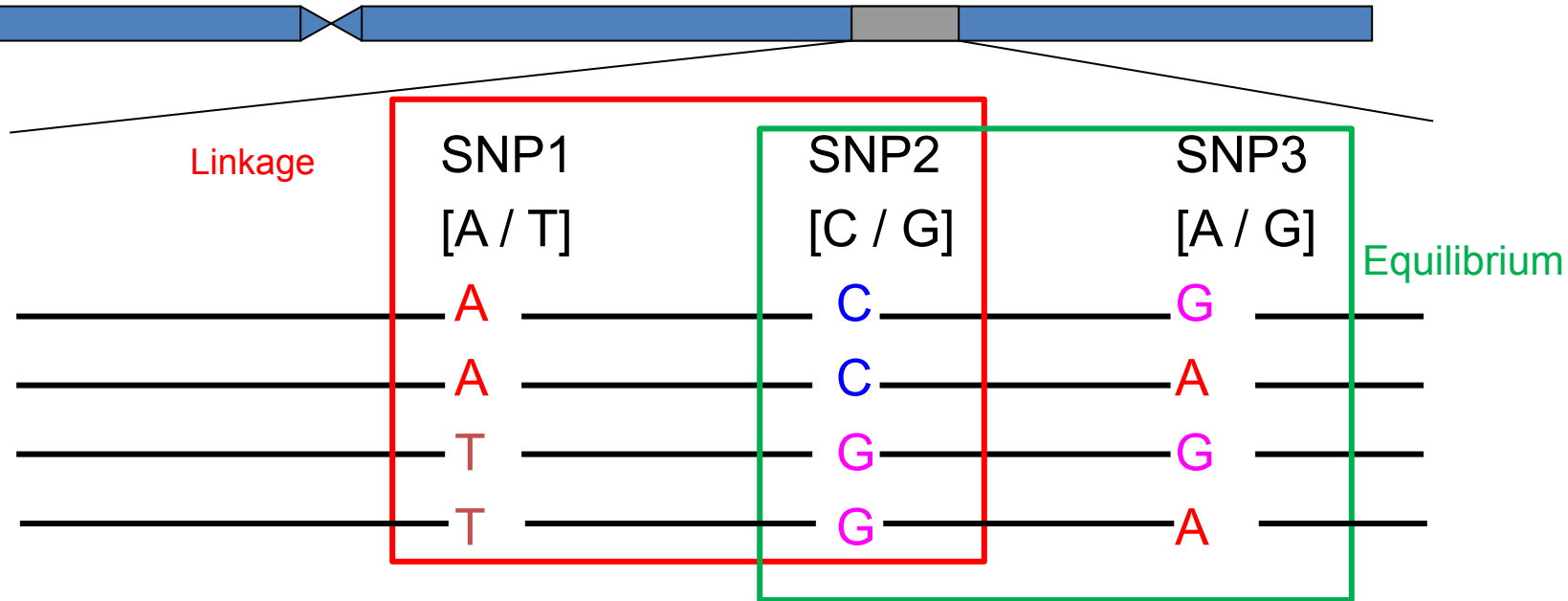
- Linkage disequilibrium
- Linear regression



Linkage disequilibrium recap

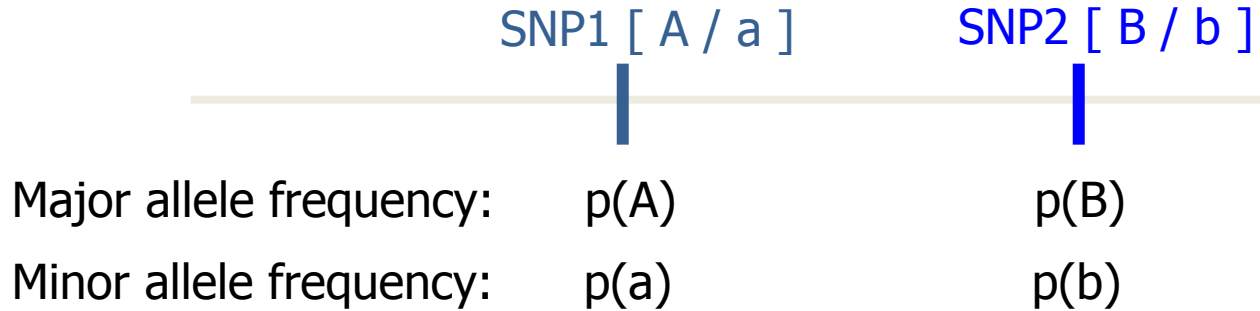


Linkage disequilibrium (**LD**)



Haplotype: specific combination of alleles that appear in the same chromosome or segment (*in-cis*) and are inherited together from a single parental.

Linkage disequilibrium (**LD**)



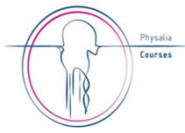
Marker segregation:

Haplotype frequency $p(ab) = p(a) \times p(b)$ Linkage equilibrium

Haplotype frequency $p(ab) \neq p(a) \times p(b)$ Linkage disequilibrium



Linkage disequilibrium (**LD**)



		SNP2 Allele		
		B	b	
SNP1 Allele	A	AB $p(A)p(B)$	$p(A)p(b)$ Ab	$p(A)$
	a	aB $p(a)p(B)$	$p(a)p(b)$ ab	$p(a)$
		$p(B)$	$p(b)$	



Example:

$p(ab) = p(a)p(b)$ **Expected Haplotype Frequencies** Linkage equilibrium

$$p(A)p(B) + p(a)p(B) = p(B)\{p(A) + p(a)\} = p(B)$$



Linkage disequilibrium (**LD**)

		SNP2 Allele			
		B	b		
SNP1 Allele	A	$p(A)p(B)+D$	$p(A)p(b)-D$	$p(A)$	
	a	$p(a)p(B)-D$	$p(a)p(b)+D$	$p(a)$	
		$p(B)$	$p(b)$		

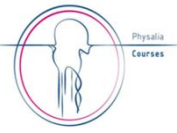
Example:

$$p(ab) \neq p(a)p(b) \rightarrow p(ab) = p(a)p(b) + D$$

**Non-Expected Haplotype
Frequencies (*D* is LD degree)**

$$\{p(A)p(B)+D\} + \{p(a)p(B)-D\} = p(B)\{p(A)+p(a)\} = p(B)$$

Linkage disequilibrium (LD)



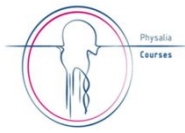
Question: What markers are in LD?

- **A** frequency=0.6; **B** frequency=0.3; haplotype **ab** frequency =0.40
- **a** frequency=0.4; **C** frequency=0.3; haplotype **AC** frequency=0.18
- haplotype **bC** frequency=0.49



Linkage disequilibrium (**LD**)

Most common measurements



$$-1 \longleftarrow D'_{\text{(Lewontin, 1964)}} \longrightarrow 1$$

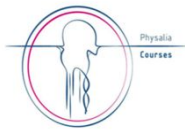
Recombination

$$0 \longleftarrow r^2 \longrightarrow 1$$

Correlation



Linkage disequilibrium (**LD**)



D' (Lewontin, 1964)

$$D' = D / D_{\max}$$

$$D_{\max} = \min (p(A)p(B), p(a)p(b)) \quad D < 0$$

$$D_{\max} = \min (p(A)p(b), p(a)p(B)) \quad D > 0$$

D' sign depends on the reference allele

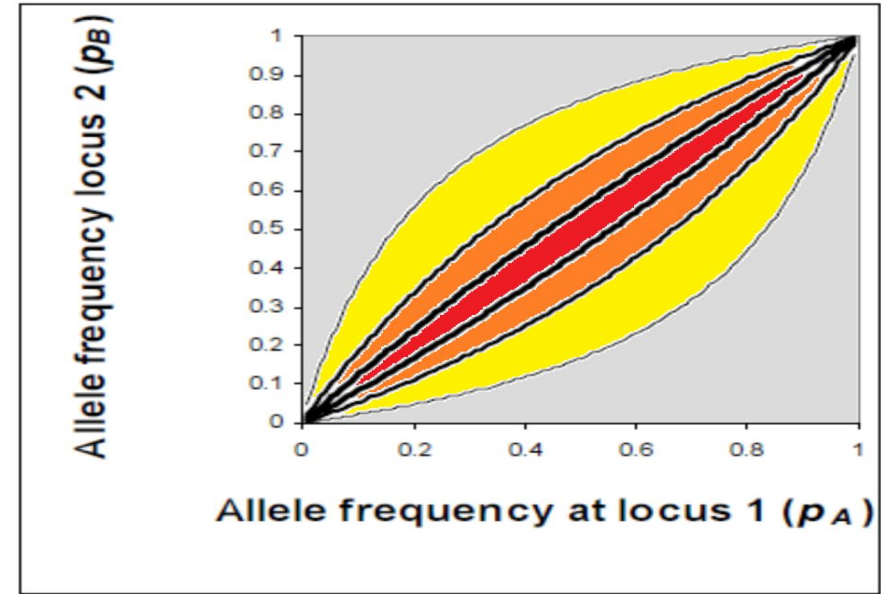


Linkage disequilibrium (**LD**)

r^2 (Hill & Robertson, 1968)

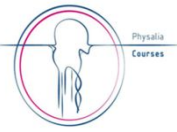
$$r^2 = \frac{D^2}{p(a)p(b)p(A)p(B)}$$

- Between 0 - 1
- Depends on allele frequency
- Most common metric



$r^2 > 0.80$
 $r^2 > 0.50$
 $r^2 > 0.20$
 $r^2 < 0.20$

Linkage disequilibrium (**LD**)



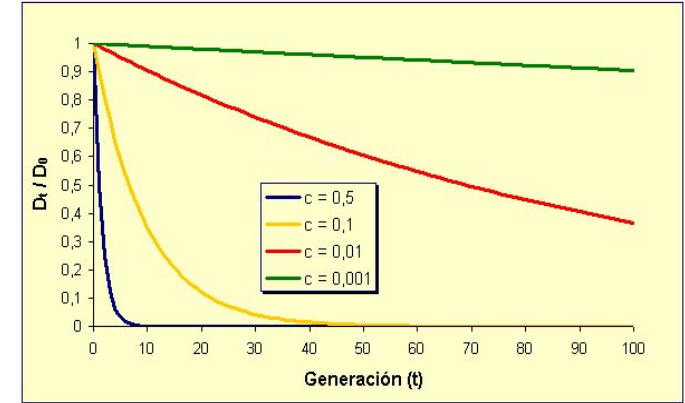
- Non-random allele segregation between 2 or more loci
- Loci on the same chromosome with LD have larger probability of being transmitted together (less likely recombination)
- Physically close loci tend to be in higher LD, as recombination rates are low



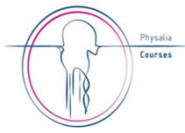
Linkage disequilibrium (**LD**)

Possible causes of LD

- Close location
- Low recombination rates
- Recent populations with low founder size
- Selective breeding (e.g. Holstein Long range LD)
- New mutations
- Inbreeding
- Population bottlenecks
- Population stratification
- Asexual reproduction



Linkage disequilibrium (**LD**)

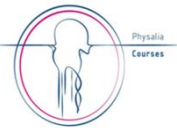


Applications

- We can infer the genotype of one locus given we know the genotype of another locus with LD (Imputation)
- We can predict the effect of a gene, given we know a marker genotype in LD (GWAS – GWP)
- Caution
 - LD may disappear in future generations due to new recombinations. Consider this cautiously when applying results to further apart generations or low LD markers
 - Differentiate between statistical linkage and physical linkage



Linkage disequilibrium (**LD**)



Why do we care about LD?

The power of an indirect association study depends on 4 parameters:

- Disease-allele effect size
- Disease-allele frequency
- Marker-allele frequency
- Extent of LD between marker and disease locus



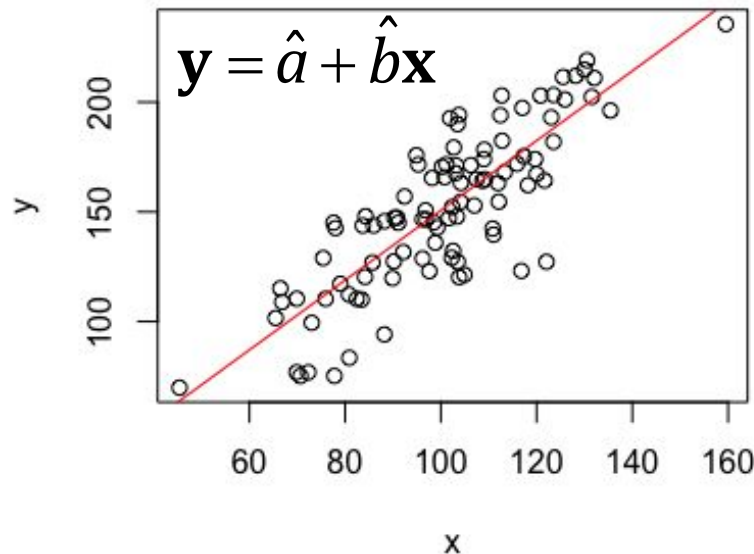
Linear regression recap



Linear regression

$$\mathbf{y} = a + b\mathbf{x} + \mathbf{e}$$

- Estimate the line that minimizes the MSE $\text{Sum}(y - \hat{y})^2$.



$$\hat{b} = \text{cov}(\mathbf{x}, \mathbf{y}) / \text{Var}(\mathbf{x})$$

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

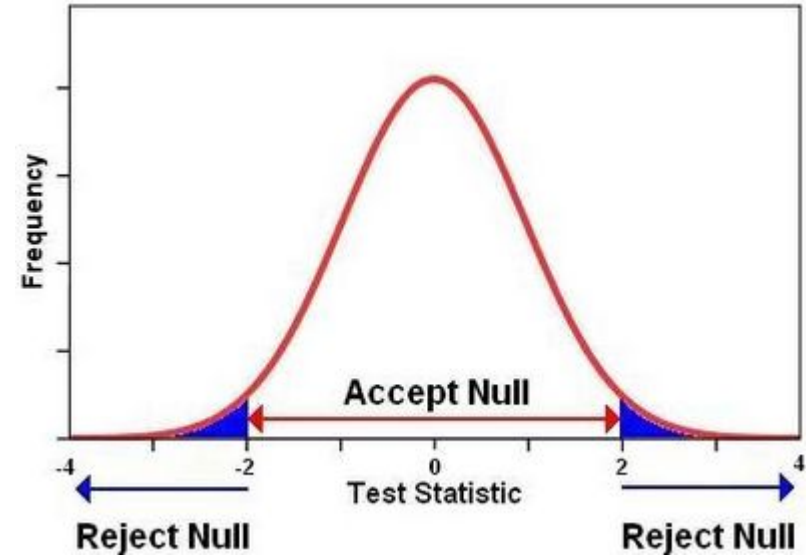
Is the b estimate statistically significant? → Real effect (association)

Linear regression – Hypothesis test

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

- Continuous variable → t-Student
- Binary variable → Chi-squared, log-ratio test



Significance level α (e.g. 0.05, 0.01,...)

Linear regression – Hypothesis test

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

	One-tail (left)	Two-tails	One-tail (right)
Contrast	$H_0 : \beta_1 = b_1$ $H_1 : \beta_1 < b_1$	$H_0 : \beta_1 = b_1$ $H_1 : \beta_1 \neq b_1$	$H_0 : \beta_1 = b_1$ $H_1 : \beta_1 > b_1$
Statistic (t-Student)	$t = \frac{\hat{\beta}_1 - b_1}{\sqrt{\hat{s}_R^2 / SS_{xx}}} \text{ con } \hat{s}_R^2 = \frac{SS_{yy} - \hat{\beta}_1 SS_{xy}}{n-2}$		
Reject Region	$t < t_{\alpha, n-2}$	$ t > t_{1-\alpha/2, n-2}$	$t > t_{1-\alpha, n-2}$

$$SS_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y},$$

$$SS_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 = n\sigma_x^2$$

$$\hat{s}_R^2 = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n e_i^2}{n-2} = \frac{SS_{yy} - \hat{\beta}_1 SS_{xy}}{n-2} \quad SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n y_i - (\beta_0 + \beta_1 x_i)^2$$

$$t = \frac{\hat{\beta}_1 - b_1}{\sqrt{\hat{s}_R^2 / SS_{xx}}}$$

Does my *t*-statistic value belong to the null hypothesis distribution

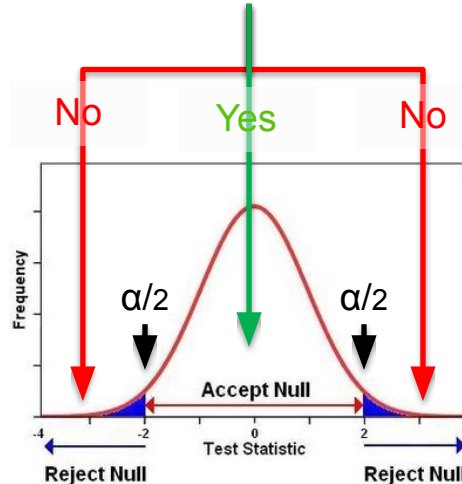


$$H_1 : \beta_1 \neq 0$$

$$t = \frac{\hat{\beta}_1 - b_1}{\sqrt{\hat{s}_R^2 / SS_{xx}}}$$


$$2 \cdot (1 - p(z \leq t))$$

If $p\text{-value} < \alpha \rightarrow$ **reject** null-hypothesis

[illegible]

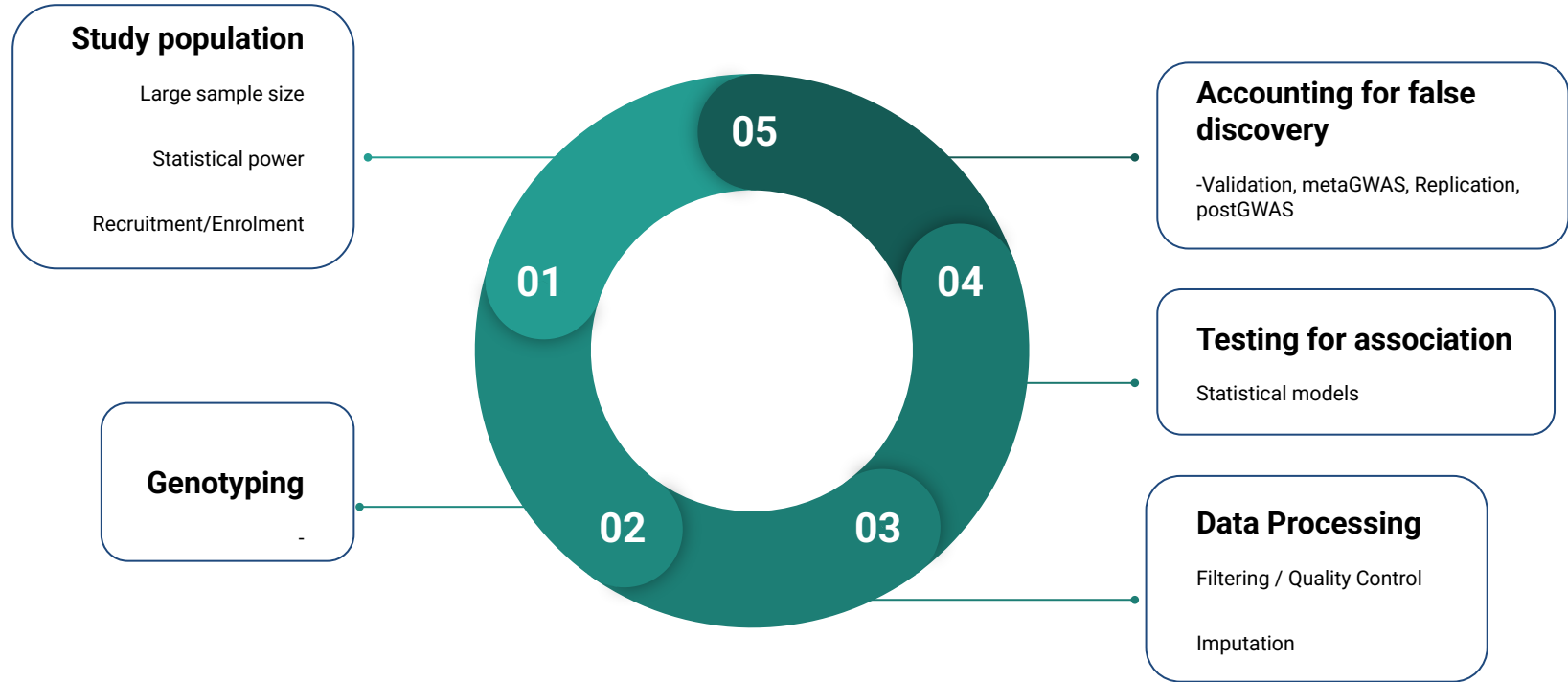
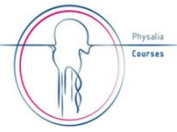
Additional material

Compute the t-student statistic given some data, and accept or reject the null-hypothesis

- 1.Basis_of_linear_regression.R
- 2.Basis_of_logistic_regression.Rmd



conduction a GWAS



Genome-wide association studies (**GWAS**)

association is one type of statistical problem

- discovery of interesting relationships among variables in large data sets (i.e., **association**);
- division of data sets into several discrete groups (i.e., **clustering**);
- assignment of observations to groups (i.e., **classification**);
- Extrapolate quantitative outputs based on attributes of observational units (i.e., **prediction**);
- etc.



Inference vs Prediction

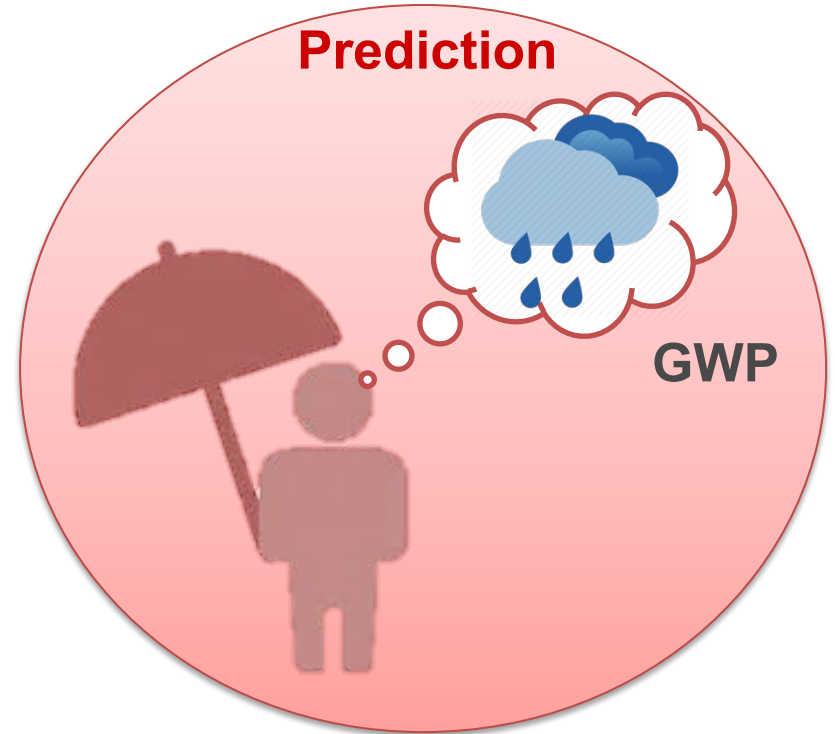
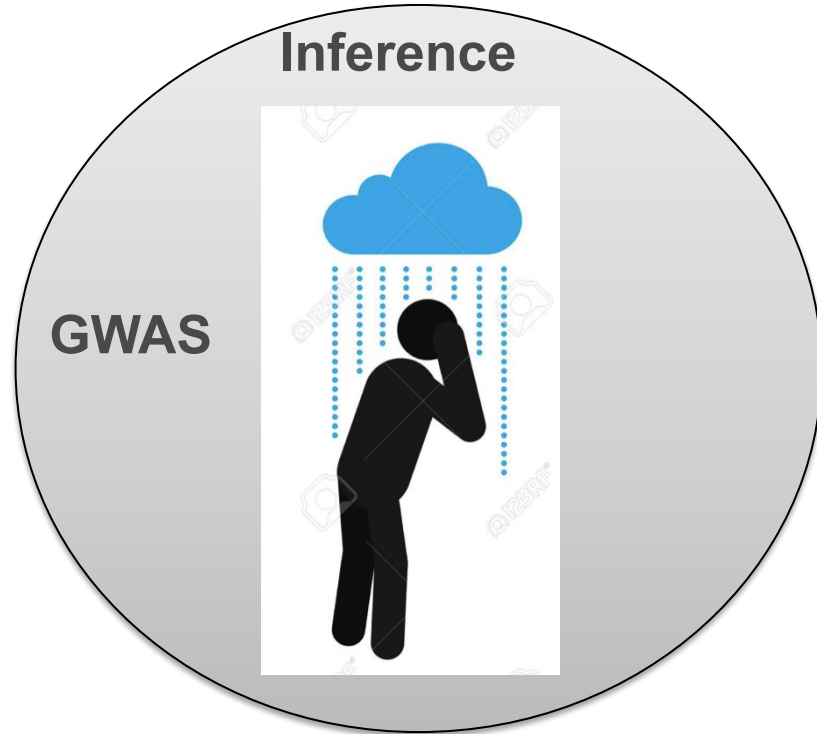
Inference

- Determine the effect of a covariate on the response
- Determine the causal relationship between a covariate and the response (Need specific experimental designs)

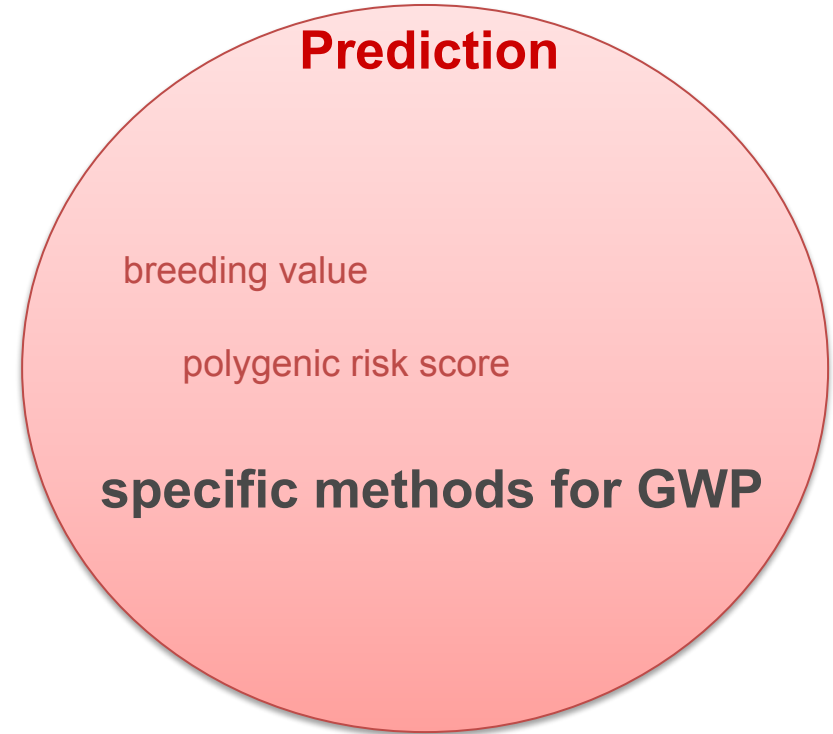
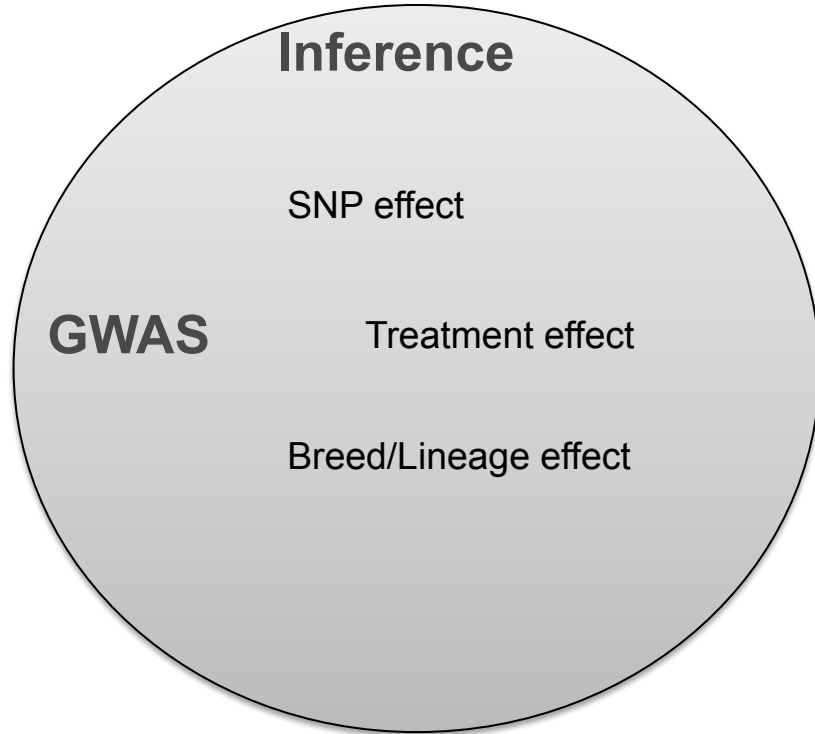
Prediction

- Educated guess of the outcome
- Expected behaviour in the future
- Based on proxies/markers

Inference vs Prediction



Inference vs Prediction



Genetic basis of GWAS-GWP

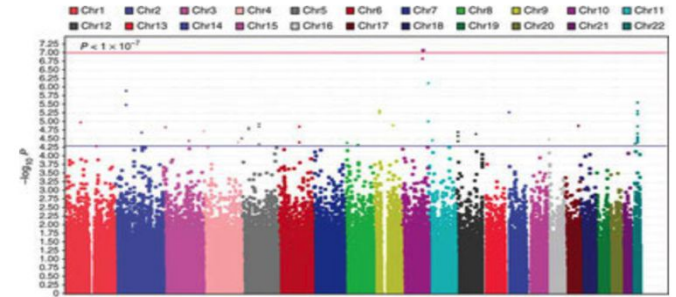
GWAS goal

- Detect genomic markers/regions associated to phenotypes (traits) of interest
- Find biological pathways of interest
- Interaction between treatments/drugs and genes

GWP goal

- Calculate the genomic risk score /predisposition of a disease or trait
- Calculate the genomic merit of individuals
- Predict future performance
- Sire/dam selection in animal breeding
- Use methods that apply shrinkage on inference

Some examples



GWAS – Rheumatoid arthritis

BMC Proceedings

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Volume 3 Supplement 7

Genetic Analysis Workshop 16

Proceedings | [Open Access](#) | [Published: 15 December 2009](#)

Detecting single-nucleotide polymorphism by single-nucleotide polymorphism interactions in rheumatoid arthritis using a two-step approach with machine learning and a Bayesian threshold least absolute shrinkage and selection operator (LASSO) model

[Oscar González-Recio](#), [Evangelina López de Maturana](#), [Andrés T Vega](#), [Corinne D Engelman](#) ✉ & [Karl W Broman](#)

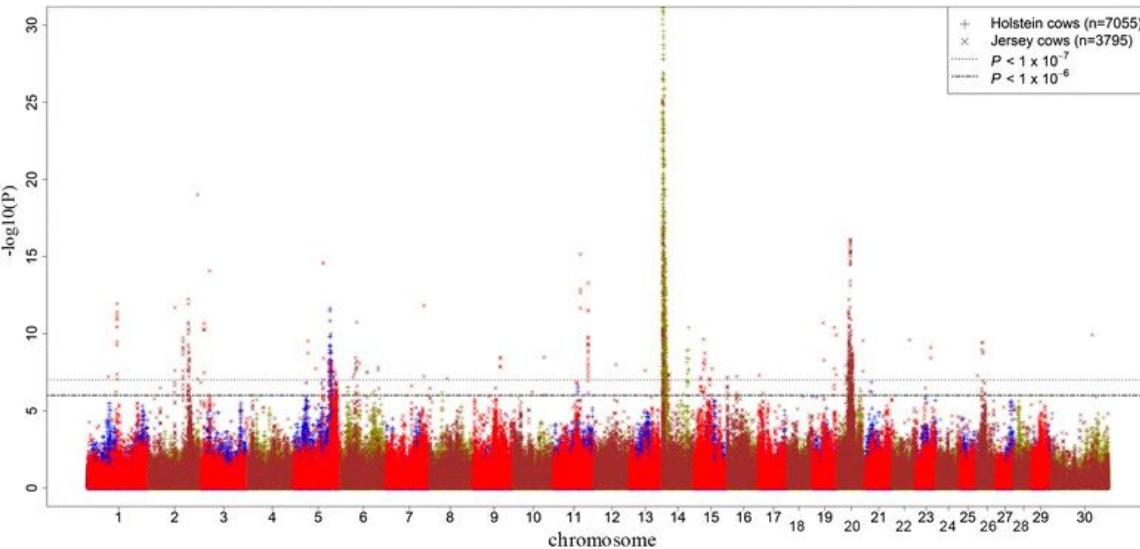
[BMC Proceedings](#) 3, Article number: S63 (2009) | [Cite this article](#)

504 Accesses | 4 Citations

Chrom:		6-HLA REGION						6	1	2	2	6	8	8	8	9	10	10	12	12	13	14	14	16	17	18	18	19	21
GENE		Class II	unknown	LY6G8D	unknown	unknown	unknown	unknown	PTPN22	unknown	CD28	unknown	ZFPM2	PSD3	unknown	WDFY4	unknown	PDE1B	unknown	unknown	RIN3	unknown	GRIN2A	WNT3	DSC3	unknown	unknown	unknown	unknown
SNP		rs10484560	rs2395175	rs3749952	rs3763338	rs660895	rs9262632	rs9369650	rs2476601	rs2353317	rs3181096	rs10094729	rs10103119	rs1038848	rs10976357	rs2671682	rs4759005	rs1022232	rs2365675	rs26543	rs12885166	rs234592	rs1875206	rs10514911	rs12455854	rs1389699	rs8104309	rs1041778	
HLA REGION	rs10484560			G-C					G-G						G-C			G-T											
	rs2395175								G-G						G-C			G-T											
	rs3749952									T-A	T-C	T-A									T-A								
	rs3763338																												
	rs660895								A-G																				
	rs9262632												A-A																
Major effect		A			G		T	A			A	A	G		A	T				T	T	C						T	
Previously associated to RA		Yes	Unknown	Yes	Unknown	Unknown	Unknown	Unknown	Yes	Unknown	Yes	Unknown	No	No	Unknown	No	Unknown	No	Unknown	Unknown	Unknown	Unknown	No	Yes	No	Unknown	Unknown	Unknown	Unknown

RISK FACTOR
Protective factor

GWAS – Milk yield dairy cattle



- **DGAT1** gene on BTA 14 in dairy cows (HOL and JER)
- milk fat content

BMC Genetics

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Research article | [Open Access](#) | Published: 22 July 2015

Validation of markers with non-additive effects on milk yield and fertility in Holstein and Jersey cows

Hassan Aliloo , Jennie E. Pryce, Oscar González-Recio, Benjamin G. Cocks & Ben J. Hayes

BMC Genetics 16, Article number: 89 (2015) | [Cite this article](#)

4154 Accesses | 8 Citations | 3 Altmetric | [Metrics](#)



GWAS – Fertility dairy cattle

A number of **fertility-related** traits
(additive and dominance)

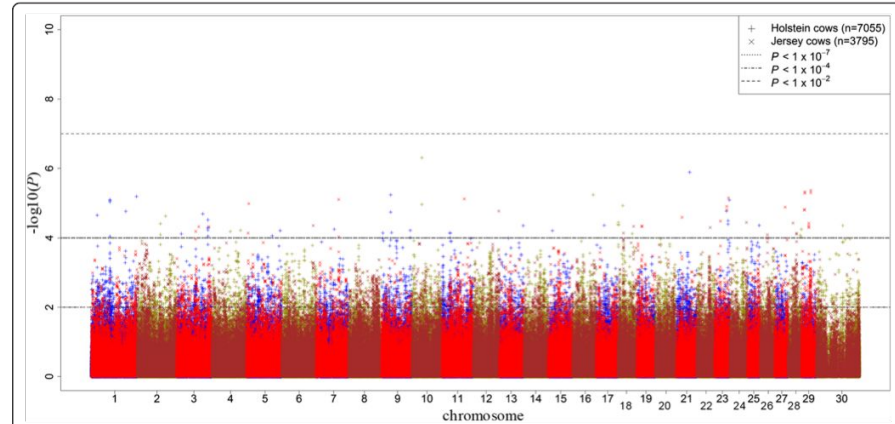


Fig. 4 Distribution of dominance SNP effects for fertility. Manhattan plot of all dominance SNP effects for calving interval in discovery and validation populations with chromosome number on horizontal axis and $-\log_{10}(P\text{-value})$ on vertical axis

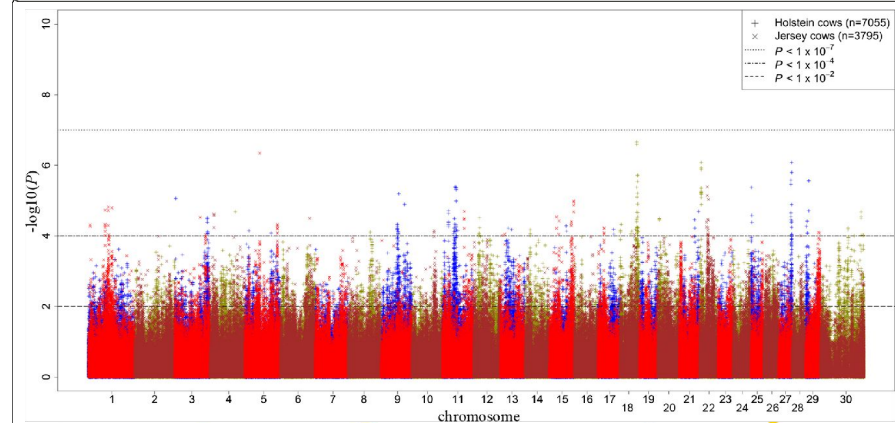


Fig. 2 Distribution of additive SNP effects for fertility. Manhattan plot of all additive SNP effects for calving interval in discovery and validation populations with chromosome number on horizontal axis and $-\log_{10}(P\text{-value})$ on vertical axis

GWAS – curly hair in cattle

naturegenetics

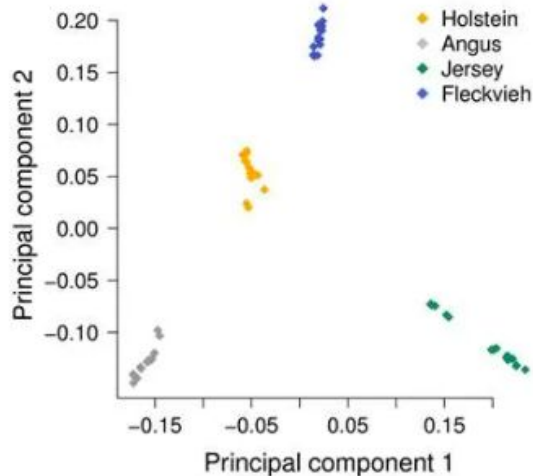
Article | Published: 13 July 2014

Whole-genome sequencing of 234 bulls facilitates mapping of monogenic and complex traits in cattle

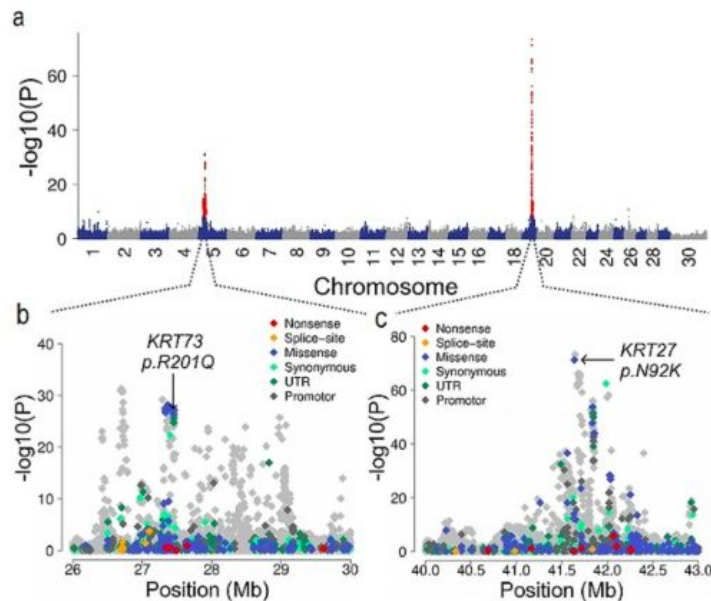
Hans D Daetwyler, Aurélien Capitan, [...] Ben J Hayes

Nature Genetics 46, 858–865(2014) | Cite this article

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From: Whole-genome sequencing of 234 bulls facilitates mapping of monogenic and complex traits in cattle



Manhattan plot showing the association of 17,640,970 imputed variants with the proportion of daughters with curly hair in 3222 Fleckvieh bulls (a). Red dots represent variants with $P < 10^{-9}$. Detailed overview of the associated regions on chromosomes 5 (b) and 19 (c). Variants in the promoter (defined to encompass 1,000 bp upstream of the transcription start), in the untranslated regions (UTR) and in the amino acid coding region are highlighted with different color. The associated region on BTA5 encompasses Krt71, which underlies curly hair in various species. Variant calling yielded four missense mutations in Krt71 (p.R133W, p.F143I, p.N177I, p.P452H); however, none of them was polymorphic in the 43 resequenced Fleckvieh animals. Functional annotation of the variants within the QTL region on BTA5 revealed that 12 closely linked missense mutations in Krt73, Krt2 and Krt76 are highly significantly associated with curly hair in Fleckvieh cattle. Among those, only the p.R201Q mutation in Krt73 (c.G602A, chr. 5: 27,445,800 bp, ss682156288) was predicted to be damaging by PolyPhen-2 and SIFT analysis.

GWAS - Schizophrenia

Molecular
Psychiatry

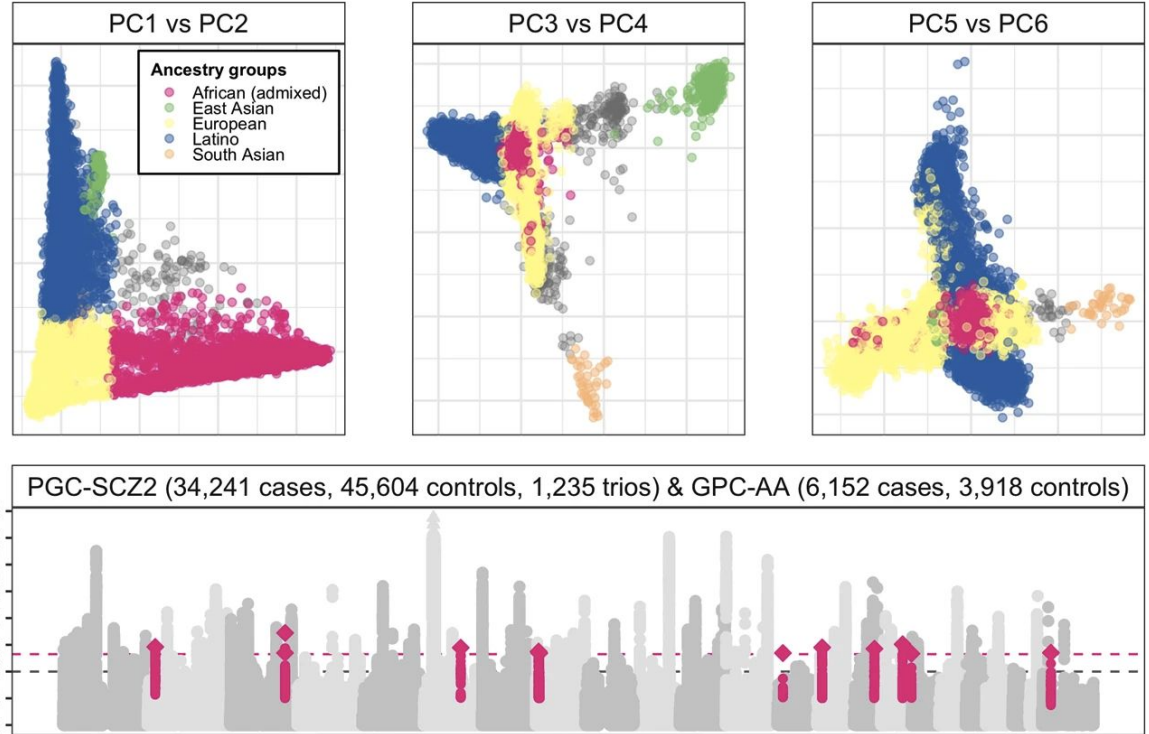
Article | [Open Access](#) | Published: 07 October 2019

Contributions of common genetic variants to risk of schizophrenia among individuals of African and Latino ancestry

Tim B. Bigdeli , Giulio Genovese, [...] Carlos N. Pato

Molecular Psychiatry (2019) | [Cite this article](#)

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GWAS - Covid-19

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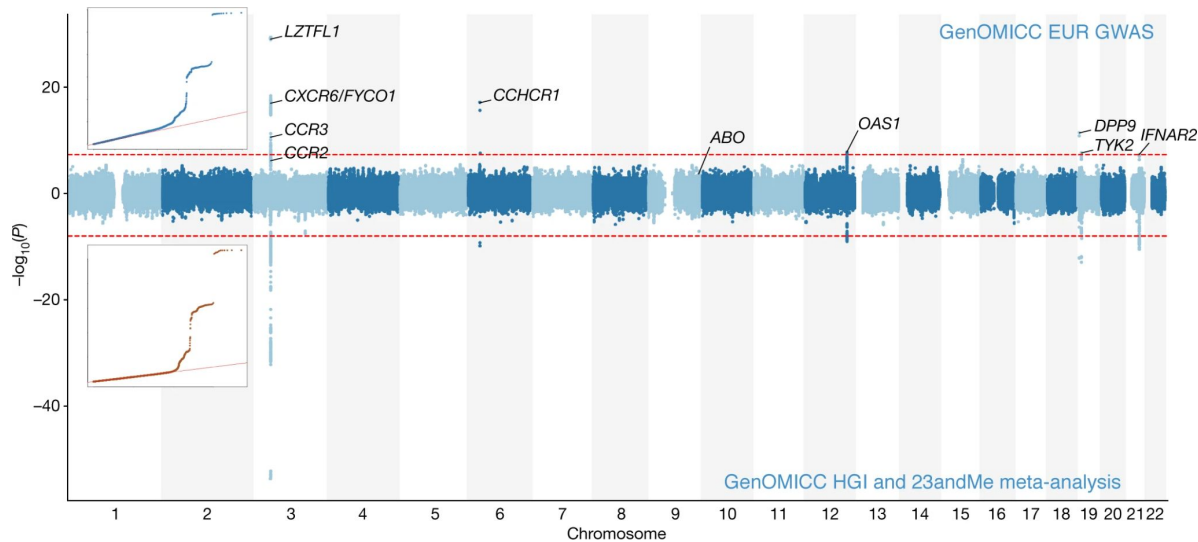
Article | Published: 11 December 2020

Genetic mechanisms of critical illness in COVID-19

Erola Pairo-Castineira, Sara Clohisey, [...]J. Kenneth Baillie [✉](#)

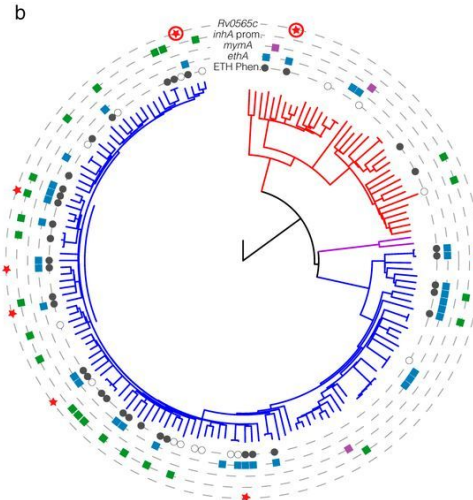
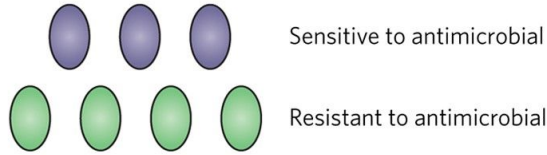
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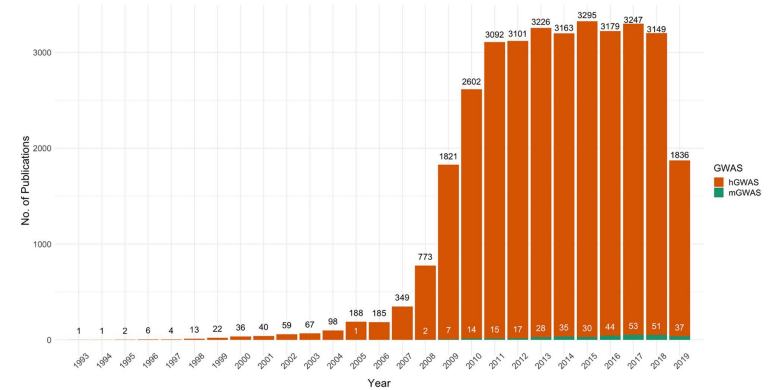


GWAS - bacterial

GWAS in bacteria



Not as popular as human GWAS (hGWAS)



Boosted by WholeGenomeSequencing of bacterial genomes along with phenotypes (e.g. antimicrobial resistance).

GWAS - **bacterial**

Particularities

- Haploid organisms
- Strong population structure (Difficult to find Causal mutation and epistatic effects)
- Great variation of the pan-genome
- Low sample sizes
- Unavailability of SNP-chips (necessary whole-genome sequencing)

Relevant in

- Antibiotic susceptibility
- Susceptibility to disinfectants
- Transmissibility
- Disease presentation/severity
- Carriage duration
- Invasiveness



Doing a GWAS is “easy”



Doing a GWAS is “easy”



Doing a GWAS CORRECTLY is not



What will you **learn**

- How GWAS work
- Use the right type of analyses
- Identify and understand the individual steps involved in a GWAS project
- Understand the limitations of GWAS
- Visualize results of GWAS
- Assemble the different steps into a reproducible pipeline



Regarding GWAS - **what we will do**

0. getting the data
1. data preprocessing (**EDA**)
2. data preprocessing (**filtering**)
3. **imputation** of missing genotype data
4. **GWAS** basic models
5. single SNP vs many SNP
6. continuous/binary traits
7. population structure
8. Manhattan plots/qq-plots (post-hoc analysis)
9. build the **pipeline**

